Adaptive Receptive Field Network for Small Objects Detection in Remote Sensing Images

Jinlong Zhang, Sanping Zhou, *Member, IEEE*, Ye Deng, *Member, IEEE*, and Le Wang, *Senior Member, IEEE*

***Abstract*—Small object detection in remote sensing images is severely hampered by the significant scale variation even among small objects. Conventional methods often rely on a static receptive field to incorporate context, leading to a critical scale mismatch: oversized fields submerge tiny objects in background noise, while undersized fields fail to capture sufficient context for larger ones. To resolve this dilemma, we propose the adaptive receptive field network based on the mixture of experts (ARFN-MoEs), a novel architecture that dynamically tailors receptive fields to specific object scales. Our framework is built upon a set of scale-specialized experts, each expert module with a distinct receptive field optimized for a particular scale range. To orchestrate these experts, we introduce a grid-level router that intelligently dispatches local feature regions to the most appropriate expert, ensuring a precisely scale-matched contextual feature extraction. Furthermore, a grid-level balance loss is designed to encourage the uniform utilization of all experts during training, preventing model degradation. Extensive experiments on multiple public benchmark datasets demonstrate the effectiveness and superiority of our proposed method for small object detection in remote sensing images. The source code will be made available at: https://github.com/GDragon986/ARFN-MoE**

***Index Terms*—Dynamic receptive field, mixture of experts (MoEs), remote sensing images, small object detection.**

I. Introduction

OBJECT detection in remote sensing images is a fundamental task for various applications, such as urban planning and national security [1], [2]. Due to the unique imaging process, the images often contain a multitude of small objects (typically defined as having a pixel area below 32×32 [3]), as illustrated in Fig. 1. Small objects typically contain very few pixels and therefore carry limited intrinsic information compared to medium or large objects. They often lack the fine-grained features that are essential for accurate detection. As a result, the performance of general object detectors on small objects has been consistently unsatisfactory [4], [5]. Furthermore, the detection of small objects is made more difficult by the highly complex backgrounds in remote sensing images [6]. Excessive background information and noise can easily overwhelm the weak feature of small objects, making it difficult to distinguish them from the surrounding background [7]. Consequently, these factors limit the accuracy of remote sensing small object detection.

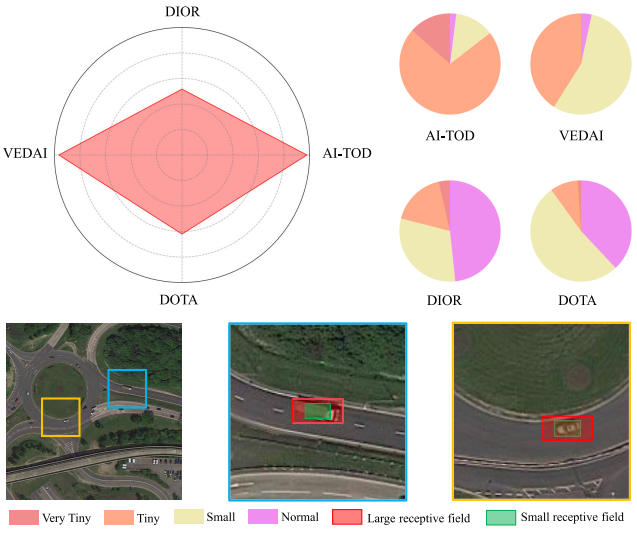


Fig. 1. Object scale distribution and variation in remote sensing datasets. (Top) Percentage of small objects in common benchmarks. (Bottom) Visual comparison illustrating the scale differences among various small objects.

In recent years, numerous studies have focused on leveraging contextual information to compensate for the limited intrinsic features of small objects, significantly improving detection performance [8], [9], [10]. However, a major limitation lies in their use of a uniform context extraction strategy for all small objects, without accounting for their individual scale variations. This "one-size-fits-all" strategy overlooks the critical fact that objects of varying sizes require different levels of contextual information to achieve optimal detection performance. For extremely small objects, the fixed contextual region is often too large, causing their weak features to be overwhelmed or interfered with excessive background information or noise. In contrast, for relatively larger small objects, the same contextual region may be too limited, failing to capture sufficient surrounding information to make their features distinguishable, as shown in Fig. 1. This mismatch between a

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fixed strategy and varying object sizes is a primary reason why these methods struggle to achieve the best possible detection results. Therefore, there is an urgent need for an adaptive mechanism that can dynamically adjust the contextual feature extraction process to the specific scale of each small object.

To address the aforementioned issues, this article proposes the adaptive receptive field network based on the mixture of experts (ARFN-MoEs), a novel architecture that dynamically tailors contextual feature extraction to specific object scales. The core of our method consists of three key components. First, we design a set of scale-specialized experts, each with a distinct receptive field optimized to capture contextual information for a particular range of object sizes. This design allows each expert to effectively extract the most useful contextual information for small objects within its designated scale range. Second, we introduce a grid-level router specifically designed for small objects of varying scales. This mechanism operates by dispatching the feature map into a grid and then adaptively routing features from different grid cells to the corresponding experts. Finally, to ensure all experts are utilized effectively during training, we design a grid-level balance loss to promote balanced routing decisions. By synergistically combining these components, ARFN-MoE achieves a more discriminative feature representation for small objects by capturing optimal context while mitigating interference from background noise. Extensive experiments on public benchmark datasets convincingly demonstrate the effectiveness of our approach. The main contributions of this article are summarized as follows.

1) We propose a novel ARFE-MoE, which employs multiple specialized experts with distinct receptive fields to handle the significant scale variation among small objects.
2) A grid-level router is introduced to intelligently and adaptively dispatch features from different spatial locations on the feature map to the most appropriate scale-specific expert.
3) To ensure that all experts are effectively trained and utilized, we design a grid-level balance loss, which promotes a balanced workload among the experts and prevents model degradation.

The rest of this article is organized as follows. Section I reviews the related work on small object detection and the mixture of experts (MoEs). Section III details the proposed ARFN-MoE, including its architecture and core mechanisms. Section IV presents extensive experiments and provides a thorough analysis of the results. Finally, Section V concludes this article.

II. Related Work

A. Small Object Detection

A primary strategy in small object detection involves enhancing multiscale feature representations, inspired by the success of general-purpose detectors [11]. FPN [12] established a foundational paradigm for this approach. In a typical FPN, a feature map from a higher level is fused with one from a lower level through a top–down pathway, which can be formulated as

$$\mathbf{M}_i = \mathrm{Conv}_{1\times1}(\mathbf{F}_i) + \mathrm{Upsample}(\mathbf{M}_{i+1}) \quad (1)$$

where $\mathbf{F}_i$ represents the feature map from the backbone's ith stage, $\mathbf{M}_{i+1}$ is the processed map from the higher level, and $\mathrm{Conv}_{1\times1}$ denotes a lateral connection via a 1×1 convolution. Building on this principle, numerous variants have been proposed. For instance, ABNet [13] introduced an adaptive feature pyramid network to capture more discriminative features, while TMAFNet [14] designed an implicit recursive pyramid to refine multiscale disparity information. MENet [15] achieves dual enhancement at both the instance level and feature level for small objects through different stages of the detector. However, while these approaches effectively enrich feature representations, they do not fundamentally address how to extract the most relevant information for objects of varying scales. Simply fusing features does not guarantee that the resulting receptive field is optimally aligned with the object's specific size.

A parallel stream of research focuses on leveraging contextual information, which is crucial when an object's intrinsic features are scarce. Contextual information refers to implicit knowledge about a specific region surrounding an object. Context-based methods can be uniformly represented as follows:

$$\hat{y} = \mathcal{P}(\mathcal{F}(\mathbf{f}_{\mathrm{feat}}, \mathbf{f}_{\mathrm{cont}}), \Theta)) \quad (2)$$

where $\mathbf{f}_{\mathrm{feat}}$ and $\mathbf{f}_{\mathrm{cont}}$ denote the extracted features and contextual information, while $\mathcal{F}$ and $\mathcal{P}$ denote the fusion operator and the predictor, respectively. Θ refers to the learnable parameters of the network. More advanced methods integrate context more dynamically, such as the context-aware modules in SINet [16] and FFCA-YOLO [17]. By incorporating global contextual information, CFMINet [18] enhanced the detection performance, especially for small and heavily occluded objects. STDet [19] effectively utilized the contextual information of the object to assist the object in detection and improve the detection performance. ORFENet [20] employs a multireceptive field strategy to enhance the contextual information of objects. LTDNet [21] develops a lightweight network specifically dedicated to small object detection, achieving a balance between detection accuracy and lightweight design. However, a fundamental limitation plagues most static context-based methods: the "scale mismatch" problem. Their reliance on a fixed-size contextual area is inherently nonadaptive to objects of varying scales, which can introduce irrelevant noise or provide insufficient information. This inflexibility often renders the extracted context counterproductive. Our work directly addresses this critical gap in spatial context modeling by introducing an adaptive mechanism that dynamically tailors the contextual region to the scale of each object.

B. Mixture of Experts

The MoE is a powerful architectural paradigm that employs multiple specialized expert subnetworks and a router network that learns to select the most suitable experts for each input [22], [23]. The final output is typically a weighted combination of the expert outputs, where the weights are determined by the gating network. For an input x, the output y of a MoE layer can be generally expressed as

$$y = \sum_{i=1}^{N} G_i(x) \cdot E_i(x) \quad (3)$$

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where N is the number of experts, $E_i(x)$ is the output of the ith expert, and $G_i(x)$ is the router's learned weight for that expert, with the constraint that $\sum_{i=1}^{N} G_i(x) = 1$. A key breakthrough was the introduction of Sparse MoE, which activates only a sparse subset of experts (e.g., Top-k) for each input token [24]. This allows for creating models with a vast number of parameters while keeping the computational cost constant, a principle pivotal to the success of large-scale language models like the Switch transformer [27].

Recently, the MoE principle has been successfully adapted for computer vision tasks. To integrate MoE with CNNs, DeepMoE [28] innovatively reconceptualized convolutional channels as experts, using a gate for adaptive channel sparsification. In the remote sensing domain, SM3Det [29] employed a grid-level sparse MoE backbone for multimodal object detection, showcasing its potential. The core philosophy of MoE, which dynamically routes inputs to specialized modules, offers a natural and powerful paradigm for solving the central challenge in small object detection: the scale-mismatch problem. We posit that objects of different scales can be treated as distinct data patterns, each best handled by an expert with an appropriately tailored receptive field. However, to the best of our knowledge, the potential of MoE to create a truly adaptive receptive field mechanism for small object detection remains unexplored. Our work is the first to propose this, adapting the MoE framework to dynamically select the optimal contextual scale for each target.

III. Methodology

A. Overview of the Framework

A significant challenge in remote sensing is the considerable scale variation that exists even within the class of small objects. Different-sized small objects require different scopes of contextual information for optimal detection. An excessively large context region can submerge the features of very small objects with background noise, while an insufficient context may fail to adequately enhance the features of normal small objects. Most existing context-aware detection methods overlook this critical factor, leading to suboptimal performance. To address these issues and further improve small object detection, we propose a novel ARFN-MoE. The overall architecture of our proposed network is illustrated in Fig. 2.

Our proposed ARFN-MoE builds upon FFCA-YOLO as its baseline. The architecture is primarily composed of three key components: an adaptive contextual feature extraction backbone, a multiscale feature fusion neck, and a multiscale detection head. At the core of our architecture is the adaptive receptive field convolution (ARFC), embedded within the backbone. The specific differences between this backbone and the baseline backbone are shown in Table I. We replaced the first and third Conv modules in the baseline backbone with ARFC. We did not replace the fifth Conv module because it extracts deeper features, and small object information is inherently scarce in deep features. Replacing the fifth Conv module would still yield only limited small object information. After balancing performance and model parameter count, we chose to replace only the first and third Conv modules in the baseline backbone. ARFC promotes adaptive feature extraction through two components: a group of multireceptive field experts (MFEs) and a grid-level router, which is responsible for assigning features to them. Concurrently, a long-range context share expert (LCE) captures long-range dependencies for objects of all scales, which further enriches the features of small objects. Finally, to promote the effective and balanced use of our expert modules, we have formulated a grid-level balancing loss. This prevents issues such as expert underutilization and ensures the full capacity of the architecture is harnessed.

TABLE I
Detailed Structure and Parameters Counts of Baseline Backbone and ARFNet

NO.	Baseline	Input	Output	Params	ARFNet	Input	Output	Params
0	Conv	3	48	5280	Conv	3	48	5280
1	Conv	48	96	41664	ARFC	48	96	60601
2	CSP	96	96	111744	CSP	96	96	111744
3	Conv	96	192	166272	ARFC	96	192	228889
4	CSP	192	192	444672	CSP	192	192	444672
5	Conv	192	384	664320	Conv	192	384	664320
6	CSP	384	384	2512896	CSP	384	384	2512896
7	SPPF	384	384	369792	SPPF	384	384	369792

B. Adaptive Receptive Field Convolution

Inspired by the task specialization capabilities of MoE models, we introduce the ARFC to address a key challenge in small object detection: the scale-mismatch between objects and their contextual feature regions. This module is designed to adaptively capture context, preventing feature submergence for tiny objects while ensuring sufficient information for larger ones. The structure of ARFC is illustrated in Fig. 2. This module is primarily composed of two main branches: the LCE branch and the MFE branch. The LCE branch acts as a shared expert responsible for extracting long-range contextual information from small objects across all scales. Its primary role is to ensure that a sufficient amount of context is captured for subsequent processing. The MFE branch performs adaptive feature extraction by matching the scale of each small object to an optimal receptive field. This scale-adaptive mechanism ensures the extraction of highly effective and relevant contextual features. Finally, the features from both branches are fused through element-wise addition, followed by a small-kernel convolution that further integrates the information, yielding a final contextual feature map optimized for small object detection.

1) Long-Range Context Shared Experts: To capture long-range contextual information for small objects of all scales, and drawing inspiration from prior works [30], we designed an LCE. This module is incorporated into the ARFC framework, where it functions as the shared expert. LCE is specifically designed to capture global contextual information by establishing dependencies among distant pixels relevant to small objects. An illustration of LCE is presented in Fig. 2. The input feature $\mathbf{X}$ is first processed by average pooling to capture the local context feature, followed by a 1×1 convolution that halves the channel count

$$\mathbf{F}^{\text{pool}} = \text{Conv}_{1\times 1}\left(\mathcal{P}_{\text{avg}}(\mathbf{X})\right) \tag{4}$$

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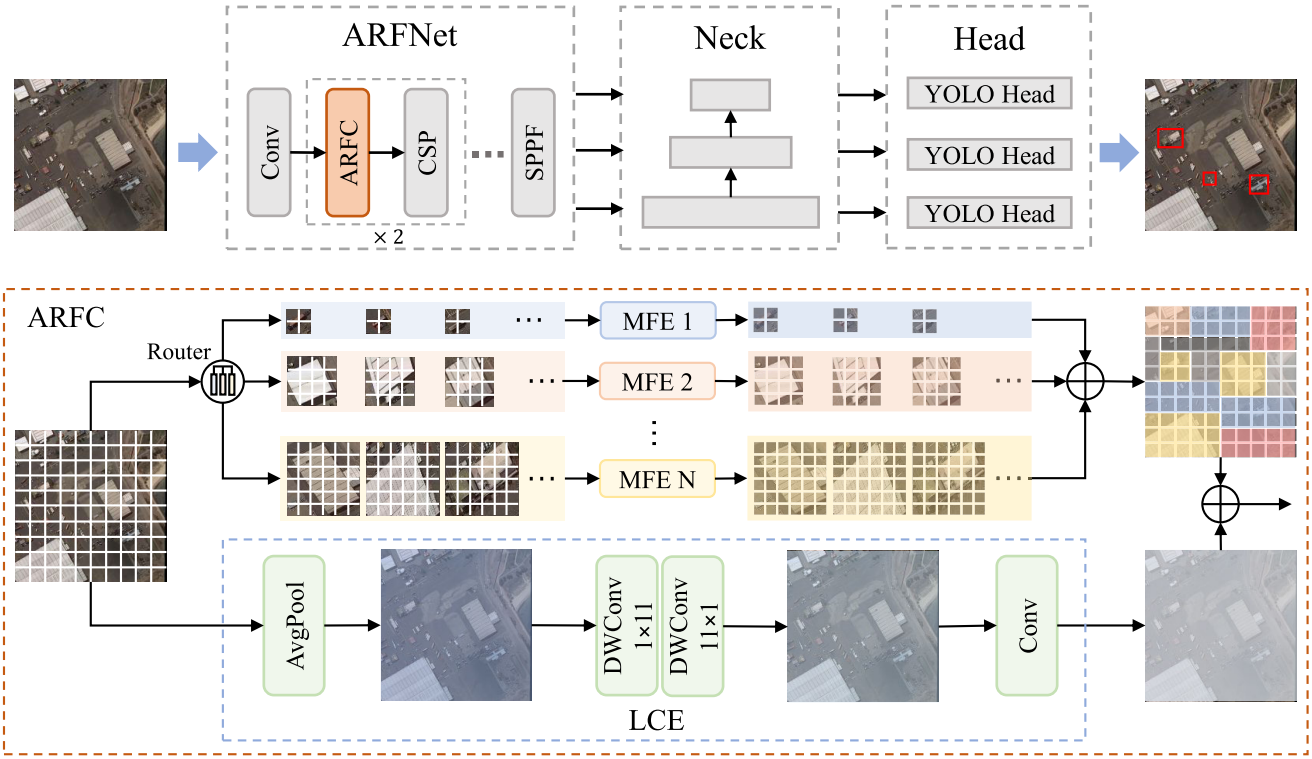


Fig. 2. Architecture of the proposed ARFN-MoE, where our ARFC module replaces a conventional convolution in the backbone. The ARFC is comprised of a grid-level router and a set of MFE.

where $\mathcal{P}_{\text{avg}}$ denotes the average pooling operation. Subsequently, the pooled feature map is processed by sequential horizontal and vertical depth-wise strip convolutions. This two-step process is designed to effectively capture long-range contextual dependencies along both spatial dimensions

$$\mathbf{F}^{w} = \text{DWConv}_{1\times k}\left(\mathbf{F}^{\text{pool}}\right) \tag{5}$$

$$\mathbf{F}^{h} = \text{DWConv}_{k\times 1}\left(\mathbf{F}^{w}\right). \tag{6}$$

The adoption of depth-wise strip convolution in this module is motivated by two primary reasons. First, depth-wise strip convolution is inherently lightweight. Compared to a $k \times k$ depth-wise convolution, a similar effective receptive field can be achieved by applying a pair of depth-wise strip convolutions (i.e., $1\times k$ and $k\times 1$). This decomposition significantly reduces the number of parameters. Second, strip convolution is particularly advantageous for feature extraction in remote sensing images. Its linear structure is well-suited for identifying the features of elongated objects, such as bridges (BRs), while also effectively capturing the long-range contextual information that is crucial for small object detection. Finally, a 1×1 convolution is applied to restore the channel dimension of the feature map

$$\mathbf{X}_S = \text{Conv}_{1\times 1}\left(\mathbf{F}^{h}\right). \tag{7}$$

This step ensures that the output features of the LCE are dimensionally compatible for the subsequent fusion with the local contextual features extracted by the other expert branches.

There are two primary reasons for employing the LCE as a shared expert. First, and most importantly, it serves to compensate for the inherent limitations of the MFE branch. While the MFE provides fine-grained adaptive context, its finite number of experts may not cover the full spectrum of scale variations. The LCE, with its ability to efficiently capture long-range dependencies, ensures that a global contextual baseline is available for all small objects, acting as a crucial complement. Second, the strip convolution mechanism within the LCE is particularly well-suited for remote sensing images, as it excels at extracting features from objects with high aspect ratios, such as BRs and ships (SHs).

2) Grid-Level Router: A single remote sensing image typically contains small objects of varying scales. Consequently, routing all features through a single expert is often suboptimal. To ensure that objects at different locations are processed by experts with receptive fields that match their scales, we design a grid-level router. As illustrated in Fig. 2, this router dynamically assigns the local feature of each grid to the best-matching expert, thereby extracting the most beneficial contextual information for small object detection.

Let $\mathbf{x}_{i,j} \in \mathbb{R}^C$ denote the input feature vector located at the ith row and jth column of the feature map, where C represents the channel dimension. The goal of the grid-level router is to determine a routing probability distribution over N experts for each local feature $\mathbf{x}_{i,j}$. To achieve this, we introduce a learnable expert embedding matrix $\mathbf{E} \in \mathbb{R}^{D\times N}$, defined as $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_N]$, where N is the total number of experts, $\mathbf{e}_n \in \mathbb{R}^D$ represents the representative embedding vector for the nth expert.

The routing process consists of three steps: feature projection, similarity measurement, and sparse selection. First, to reduce computational complexity and enhance representation,

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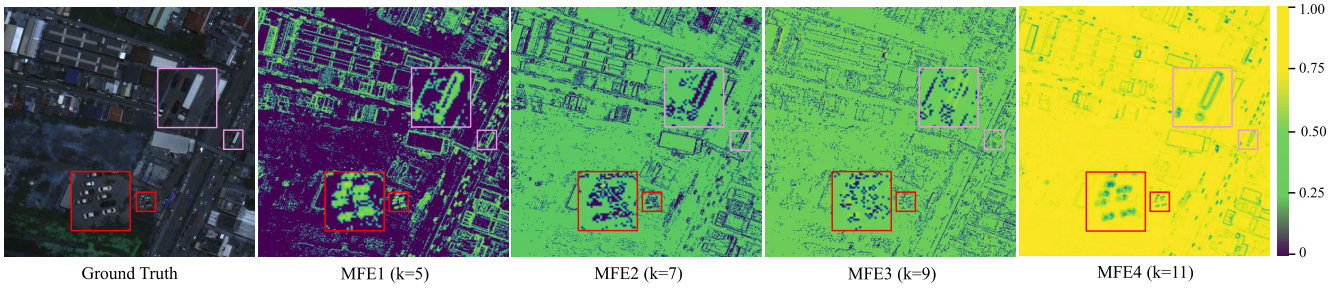


Fig. 3. Visualization of the grid-level router. The router dynamically assigns small objects of varying scales to MFE with corresponding receptive fields.

the input feature $\mathbf{x}_{i,j}$ is projected into the embedding space via a learnable transformation matrix $\mathbf{W}$. Then, the router calculates the cosine similarity between the projected feature and each expert embedding $\mathbf{e}_n$. The similarity scores between the input feature $\mathbf{x}_{i,j}$ and each expert are then normalized using the softmax function. Finally, sparse selection is performed using a Top-k strategy, which identifies the k experts with the highest similarity scores for the subsequent processing of the input feature $\mathbf{x}_{i,j}$. Then, the weights of the Top-k experts are renormalized via softmax to ensure they sum to one. The weights for the remaining $N-k$ unselected experts are explicitly set to zero. This procedure ensures that only the Top-k best-matching experts are activated for each input feature $\mathbf{x}_{i,j}$. The overall workflow of the grid-level router can be formulated as follows:

$$G\left(\mathbf{x}_{i,j}\right) = \mathrm{TOP}_k\left(\mathrm{Softmax}\left(\frac{\mathbf{W}\mathbf{x}_{i,j}\mathbf{E}}{\tau\left\|\mathbf{W}\mathbf{x}_{i,j}\right\|\|\mathbf{E}\|}\right)\right) \tag{8}$$

where τ is a scaling coefficient (temperature) that controls the sharpness of the distribution. A smaller τ encourages a sharper distribution, making the router more decisive, while a larger τ leads to a more uniform distribution. In this work, τ is set as a learnable parameter constrained between 0.5 and 1.5 to prevent training instability, with an initial value of 1.0.

Notably, the transformation matrix $\mathbf{W}$ and the expert embedding matrix $\mathbf{E}$ within the grid-level router are optimized via end-to-end training. During this process, the employment of the Top-k strategy induces sparsity in expert utilization. This mechanism progressively aligns experts possessing appropriate receptive fields with small objects of corresponding scales, while simultaneously suppressing the association between objects and experts with mismatched (excessively large or small) receptive fields. Ultimately, as illustrated in Fig. 3, this enables the grid-level router to precisely route small objects of specific sizes to experts with matching receptive fields, thereby maximizing the effectiveness of the extracted features.

The final output of the MoE layer, $\mathbf{y}(x_{i,j})$, is computed as the weighted aggregation of the selected experts' outputs

$$\mathbf{y}\left(\mathbf{x}_{i,j}\right) = \sum_{n=1}^{N} G\left(\mathbf{x}_{i,j}\right) \cdot f_E^n\left(\mathbf{x}_{i,j}\right) \tag{9}$$

where $f_E^n(\cdot)$ represents the feature extraction function of the nth expert. Through this adaptive assignment strategy, the network explicitly directs features of potential small objects to experts with appropriate receptive fields, maximizing the efficacy of feature extraction.

3) Multireceptive Field Experts: While the LCE effectively captures long-range contextual information using depth-wise strip convolutions, its reliance on only horizontal and vertical kernels restricts its ability to comprehensively capture all contextual features. To overcome this limitation and ensure a more thorough extraction, we further designed the MFE. Each expert is designed with a unique receptive field size, enabling it to thoroughly extract contextual information tailored to small objects of a specific scale.

An illustration of MFE is presented in Fig. 4. To ensure each expert possesses a distinct receptive field capable of extracting contextual information about small objects at specific scales, we designed distinct expert modules from two perspectives. First, we employ depth-wise separable convolutions with varying kernel sizes to endow modules with distinct receptive fields. The rationale for using depth-wise separable convolutions over standard convolutions lies in the fact that multiple convolutional layers increase the number of parameters. Therefore, to reduce parameter complexity, we substitute standard convolutions with depth-wise separable convolutions, which possess fewer parameters. Second, we modulate the channel counts: for experts with smaller kernels, we increase the number of channels to enhance their capacity for extracting fine-grained details. Conversely, for experts with larger kernels, we reduce the channel count. This not only forces them to focus on capturing spatial context but also reduces the overall parameter count.

The input feature $\mathbf{X}_{\mathrm{Ei}}$ of the MFE is first processed by a $k\times k$ depth-wise convolution to capture the local contextual features of small objects. This process is formulated as follows:

$$\mathbf{F}_i^k = \mathrm{DWConv}_{k\times k}\left(\mathbf{X}_{\mathrm{Ei}}\right) \tag{10}$$

where i denotes the ith MFE module. Considering that contextual features captured with large receptive fields, while aiding classification, may lack the spatial precision required for accurate regression, we augment MFE with coordinate attention, inspired by the work of [31]. The primary function of this module is to enhance the localization capabilities of the large-receptive-field experts, thereby refining the regression performance. The feature is first pooled separately along the horizontal and vertical dimensions. Subsequently, the resulting two features are concatenated

$$\mathbf{F}_i^{h+w} = \mathrm{Cat}\left(\mathcal{P}_{\mathrm{avg}}^h\left(\mathbf{F}_i^k\right), \mathrm{Reshape}\left(\mathcal{P}_{\mathrm{avg}}^w\left(\mathbf{F}_i^k\right)\right)\right) \tag{11}$$

where Cat denotes the concatenation operation and $\mathcal{P}_{\mathrm{avg}}^h$ and $\mathcal{P}_{\mathrm{avg}}^w$ denote the average pooling operations along the

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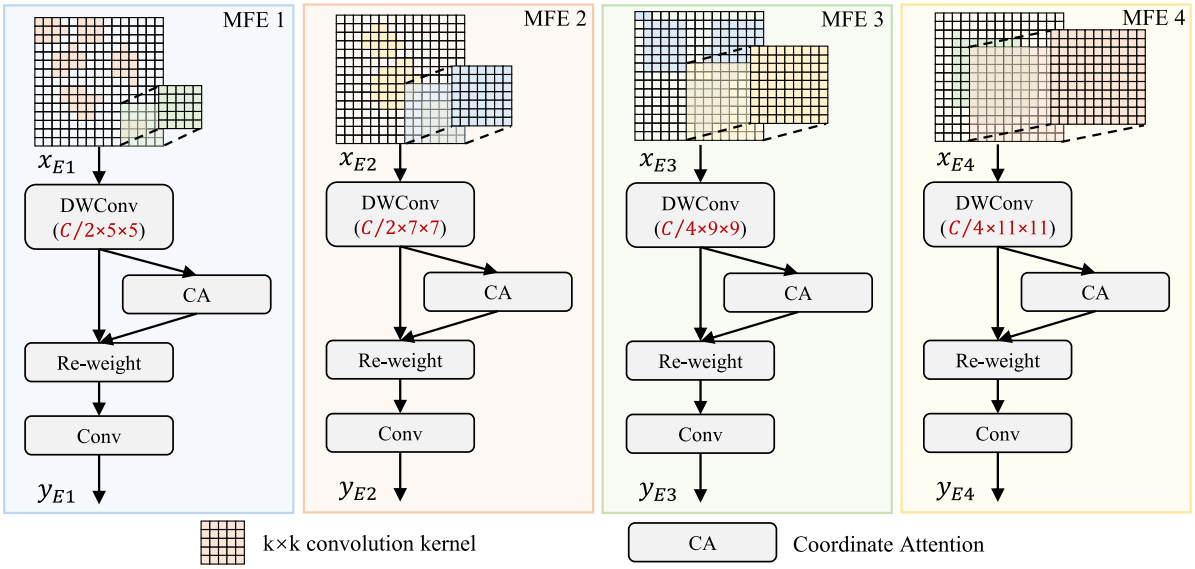


Fig. 4. Structure of the MFE. It comprises four experts with varying kernel sizes and channel numbers to extract optimal features for multiscale small objects.

horizontal and vertical axes, respectively, Reshape denotes the reshape operation. A 1×1 convolution is then applied to fuse the concatenated features, allowing information to be exchanged between the horizontal and vertical representations. Following this fusion, the features are separated again to recover the two distinct directional attention maps:

$$\mathbf{F}_i^h = \text{Split}^h\left(\text{Conv}_{1\times1}\left(\mathbf{F}_i^{h+w}\right)\right) \tag{12}$$

$$\mathbf{F}_i^w = \text{Split}^w\left(\text{Conv}_{1\times1}\left(\mathbf{F}_i^{h+w}\right)\right) \tag{13}$$

where Split^h and Split^w denote the operations that split the feature map into horizontal and vertical components, respectively.

Finally, a sigmoid activation function is applied to the processed features to generate the final coordinate attention map. This attention map is then used to reweight and enhance the original contextual features

$$\mathbf{CA}^h = \text{Sigmoid}\left(\text{Conv}_{1\times1}\left(\mathbf{F}_i^h\right)\right) \tag{14}$$

$$\mathbf{CA}^w = \text{Sigmoid}\left(\text{Conv}_{1\times1}\left(\mathbf{F}_i^w\right)\right) \tag{15}$$

$$\mathbf{y}_E = \mathbf{F}_i^k \odot \mathbf{CA}^h \odot \mathbf{CA}^w. \tag{16}$$

Here, Sigmoid function ensures that the attention map $\mathbf{CA}^h$ and $\mathbf{CA}^w$ are in range $(0, 1)$, and $\odot$ denotes the element-wise multiplication.

Multiple MFE modules, each operating with independent parameters to foster specialization, compose the second branch of the ARFC. We make these experts different from each other in two main ways: by changing the kernel size of the convolution and the number of channels. This makes sure every MFE has a different function. Specifically, MFE with small kernels has small receptive fields but more channels. This configuration allows them to focus on extracting fine-grained details and local context for smaller objects while minimizing the inclusion of background noise. Conversely, MFE with larger kernels has larger receptive fields and fewer channels. They are specialized in capturing the broader spatial context of larger small objects, thereby compensating for the limited contextual view of the small-kernel experts.

C. Loss Function

The total loss function of our proposed network, denoted as L_{total}, is composed of two main components: a detection loss L_{det}, and auxiliary grid-level balance loss L_{aux}, as formulated below

$$L_{\text{total}} = L_{\text{det}} + cL_{\text{aux}} \tag{17}$$

where c is a hyperparameter that controls the weight of the L_{aux} in the total loss. The L_{det} loss [32], which consists of the localization loss L_{box}, classification loss L_{cls}, and confidence loss L_{obj}, which can be expressed as

$$L_{\text{det}} = \sum_{i=0}^{2} a_i L_{\text{box}} + \sum_{i=0}^{2} b_i L_{\text{cls}} + \sum_{i=0}^{2} c_i L_{\text{obj}} \tag{18}$$

where i represents the feature layer used for the detector, while a_i, b_i, and c_i are the weights of the three loss functions in the different detection layers.

During the training of the MoE model, the router may develop a tendency to consistently favor a specific subset of experts. This leads to other experts being under-trained, resulting in a waste of their parameters. Furthermore, this over-reliance on a few experts can degrade the model's performance on new data and reduce its robustness. To address this issue and ensure that all the MFEs in our network are sufficiently trained, we introduce an auxiliary load balancing loss [33] that is specially designed to work together with our grid-level router. The formulation of the load balancing loss, denoted as L_{aux}, is as follows:

$$L_{\text{aux}} = \left(\frac{\sigma\left(p_{\text{mean}}\right)}{\mu\left(p_{\text{mean}}\right)}\right)^2 \tag{19}$$

where p_{mean} denotes the average probability of an expert being selected. The degree of load balance is quantified by the

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coefficient of variation (CV). A lower CV signifies that the experts are being selected more uniformly, indicating a well-balanced routing policy.

IV. Experiments

This section presents a series of experiments to validate the effectiveness of the proposed ARFN-MoE. First, the dataset used for experiments is introduced, followed by the evaluation metrics and implementation details. Subsequently, a series of comparative and ablation studies are conducted. The results of these studies are then thoroughly analyzed to rigorously validate the effectiveness of the proposed method.

A. Experimental Dataset

1) AI-TOD: AI-TOD [34] is a challenging benchmark specifically for tiny object detection in aerial images. Its distinguishing feature is the extremely small average object size of approximately 12.8 pixels. The dataset comprises 28 036 images (800×800 pixels) with 700 621 instances across eight categories: airplane (AI), BR, storage tank (ST), SH, swimming pool (SP), vehicle (VE), person (PE), and windmill (WM). Approximately, 98% of the instances are small objects. The dataset includes images of various complex scenes taken at different imaging heights. Following the MS COCO criteria, objects in the AI-TOD dataset are divided into three size ranges: [2, 8], [8, 16], and [16, 32] pixels, categorized as very tiny, tiny, and small, respectively, to facilitate the accurate evaluation of extremely small-scale objects. Following the official protocol, we train our model on the combined training (11 214 images) and validation (2804 images) sets and evaluate its performance on the test set of 14 018 images. The AI-TODv2 [5] dataset corrects certain errors and omissions in annotations based on the AI-TOD dataset, while remaining identical to the AI-TOD dataset in all other aspects.

2) VEDAI: The VE detection in aerial imagery (VEDAI) dataset [35] contains images cropped from the high-resolution Utah AGRC dataset, with a resolution of about 12.5 × 12.5 cm per pixel. This dataset contains 1246 images and is officially designed for 10-fold cross-validation. In each fold, 1089 images are used for training, while the remaining 121 images are reserved for testing. Subsequent experimental results on this dataset are all based on the average of tenfold cross-validation. Each identical scene in this dataset contains both RGB and near-infrared images, available in two resolutions: 1024 × 1024 pixels and 512 × 512 pixels. Subsequent experiments conducted on this dataset exclusively utilize the 512 × 512 RGB images. To ensure sufficient training samples per class, we exclude categories with fewer than 50 instances (e.g., plane and bus). Our evaluation focuses on the remaining eight VE-related classes: car (CA), pickup (PU), camping (CP), truck (TK), other (OT), tractor (TR), boat (BO), and van (VA).

3) DIOR: DIOR [36] is a large-scale, general-purpose benchmark for object detection in remote sensing. This dataset contains 23 463 high-resolution images at 800 × 800 pixels, annotated with 190 288 instances, of which small objects account for approximately 52%. It includes 20 categories: AI, airport, baseball field, basketball court, BR, chimney, dam, expressway service area, expressway toll station, golf field, ground track field, harbor, overpass, SH, stadium, ST, tennis court, train station, VE, and WM. For brevity, the 20 categories are hereafter referred to as C1 through C20. The dataset is characterized by a wide range of spatial resolutions (0.5 to 30 m). Following the official split, we use the 11 725 images in the trainval set for training and the 11 738 images in the test set for evaluation.

B. Evaluation Metrics

In this article, we follow the MS COCO benchmark for evaluating the detection results. Mean average precision (mAP) is used to evaluate the overall accuracy of the dataset, calculated as the mean of mAP_{50} for each category. Furthermore, we employ the $mAP_{50:95}$ metric to provide a more rigorous assessment of the model's performance on small object detection. In order to reflect the detection performance for small objects, mAP_s is used as the evaluation metric. For the AI-TOD dataset, due to its unique size distribution, stricter metrics are introduced to evaluate tiny objects. To refine the evaluation of small objects (within the size range of [0, 32]), mAP_s is further divided into three metrics based on object size in ablation experiments: mAP_{vt}, mAP_t, and mAP_s, which evaluate very tiny (vt), tiny (t), and small (s) objects, respectively.

C. Implementation Details

The proposed model was implemented in PyTorch and deployed on a workstation with an NVIDIA 3090 GPU. SGD optimizer was used with an initial learning rate 0.01, momentum 0.937, and weight decay 0.0005 to learn the parameters. The input image size was set to 800 × 800 pixels for the AI-TOD and DIOR datasets; for the VEDAI dataset, it was set to 512 × 512 pixels. The batch size during training was set to 4. For data augmentation, we employed the Mosaic technique, and all images are normalized to regulate data distribution. For the AI-TOD and VEDAI dataset, the number of epochs is set to 300. For the DIOR datasets, the number of epochs is set to 100. During the inference phase, the confidence threshold is set to 0.05. Normalized Wasserstein distance (NWD) [37] loss is added to the loss function of YOLOv5 as a supplement to the box loss.

D. Comparing State-of-the-Art Methods

1) Results on AI-TOD: To validate the effectiveness of the proposed ARFN-MoE, comprehensive experiments were conducted on the AI-TOD dataset, with results presented in Table II. The model achieves a strong overall performance of 65.3% in mAP_{50} and 28.4% in the more rigorous $mAP_{50:95}$. Compared with FFCA-YOLO, a state-of-the-art small object detection method, our proposed model achieves improvements of 2.6% and 0.8% on the mAP_{50} and $mAP_{50:95}$, respectively. Beyond the overall mAP, our proposed method also achieves the best performance on a majority of the individual object categories. Specifically, our model improves upon the next-best method's $mAP_{50:95}$ score by 2.0% on AI, 1.2% on SP,

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TABLE II

COMPARISON OF DIFFERENT METHODS ON THE AI-TOD AND AI-TODV2 DATASET. THE BEST RESULTS ARE MARKED IN BOLD. * INDICATES RESULTS ON THE AI-TODV2 DATASET

Method	AI	SH	PE	BR	SP	WM	ST	VE	mAP_{50}	$mAP_{50:95}$	mAP_{vt}	mAP_t	mAP_s	Params(M)	FLOPs(G)
Faster R-CNN [38]	22.1	19.6	4.6	5.2	7.7	0.1	19.6	13.3	27.2	11.5	0.0	8.3	24.0	41.8	134.5
RetinaNet [39]	17.6	23.4	4.1	8.7	8.0	3.1	17.3	11.3	30.8	11.7	3.4	8.8	17.3	36.3	130.1
Cascade R-CNN [40]	25.6	23.6	5.3	7.4	10.8	0.0	23.3	14.1	30.8	13.8	0.0	10.6	25.5	69.1	162.3
RepPoints [41]	15.5	28.8	6.4	8.0	4.4	0.5	26.9	20.1	34.8	13.8	2.9	12.8	18.7	36.6	117.9
DetectoRS [42]	26.1	26.5	6.2	11.4	12.0	0.1	22.9	16.0	33.1	15.1	0.1	11.3	27.6	123.2	159.7
FSANet [43]	22.2	29.8	5.9	10.8	9.4	4.2	28.2	19.6	41.4	16.3	4.4	14.6	23.4	31.9	125.2
NWD-RKA [5]	23.4	34.2	9.5	12.6	10.9	4.1	35.1	24.2	48.3	19.2	7.2	18.9	24.7	-	-
KLDet [44]	13.6	42.3	9.3	**18.7**	5.2	6.7	35.7	24.9	46.4	19.6	8.4	20.6	22.7	32.0	339.3
TBNet [45]	27.4	43.7	10.9	18.2	15.0	7.3	36.0	25.6	54.3	23.0	9.4	24.3	28.5	58.7	-
CFENet [10]	-	-	-	-	-	-	-	-	59.2	26.6	9.0	27.5	32.3	-	-
MENet [15]	27.5	42.1	10.2	16.4	18.8	8.2	37.4	24.9	56.2	23.2	9.7	23.9	25.3	58.7	148.3
STDet [19]	24.9	**46.6**	10.5	16.4	14.0	7.4	37.0	25.0	54.6	22.7	11.0	24.3	24.7	38.7	123.7
TTFNet+MGAM [47]	-	-	-	-	-	-	-	-	54.8	25.3	11.6	26.0	26.4	54.4	243.3
FFCA-YOLO [17]	36.2	41.0	16.6	18.0	20.6	9.5	45.6	33.1	62.7	27.6	10.9	31.1	33.4	**7.1**	**81.0**
ARFN-MoE (ours)	**38.2**	41.9	**16.9**	17.8	**21.8**	**11.2**	**45.7**	**33.5**	**65.3**	**28.4**	**11.9**	**31.2**	**34.2**	7.2	92.9
ORFENet* [20]	26.0	50.6	11.0	**21.1**	17.1	8.7	35.8	27.8	55.4	24.8	9.7	24.4	28.7	32.8	373.3
LTDNet* [21]	23.3	44.8	9.6	19.9	17.2	7.3	34.4	27.4	54.6	23.0	8.9	23.7	27.2	**4.9**	**38.2**
ESG_TODNet* [46]	26.6	**50.7**	11.2	20.5	16.5	8.0	35.3	27.8	55.1	24.6	9.5	24.0	29.4	32.9	387.2
ARFN-MoE*(ours)	**42.2**	42.5	**17.9**	20.5	**20.3**	**12.5**	**45.1**	**34.5**	**66.3**	**29.4**	**12.0**	**27.1**	**32.2**	7.2	92.9

and 1.7% on WM. For different object sizes, our model also achieves the best results, with $mAP_{50:95}$ scores of 11.9% for very tiny (vt), 31.2% for tiny (t), and 34.2% for small (s) objects.

We also evaluated the complexity of various methods. The proposed method possesses 7.2 M parameters and 92.9 G FLOPs, which is significantly superior to most small object detection methods. It is only marginally higher than LTDNet [21], which are currently the most lightweight approach for small object detection. This efficiency is attributed to the adoption of depth-wise separable convolutions within the MFE, a lightweight operation that prevents excessive growth in parameter count. Although the parameter count and computational complexity of our method are not the absolute lowest, it achieves a substantial improvement in small object detection performance with only a minimal increase in model complexity.

We visualize the confusion matrix on AI-TOD to intuitively display model performance, as shown in Fig. 5. In the confusion matrix, columns represent the true classes in the dataset, while rows denote the predicted classes for each sample. Data along the diagonal indicates the true positive rate, i.e., prediction accuracy. The final row of the matrix shows the false negative rate for each class, reflecting the rate of missed detections. Data along the diagonal indicates that the proposed ARFN-MoE improves prediction accuracy across all categories. For instance, accuracy for AI increased from 75% to 80%, for SP from 54% to 66%, and for WM from 70% to 78%. These target categories exhibit significant size variations within the dataset. The proposed method, through its adaptive receptive field mechanism based on a mixed expert model, selects the most suitable receptive field for small-scale targets to extract effective features, thereby enhancing prediction accuracy. The final row of data demonstrates that the proposed method simultaneously reduces the miss rate across all categories. These results further validate the effectiveness

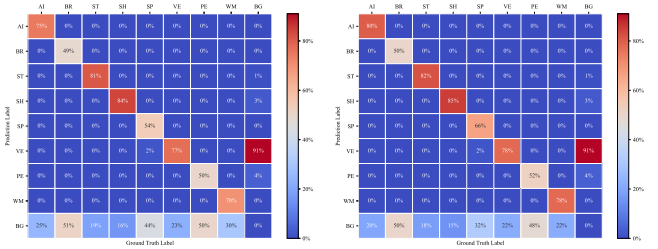


Fig. 5. Comparison of the confusion matrix between our ARFN-MoE and FFCA-YOLO on the AI-TOD dataset. (a) FFCA-YOLO. (b) Ours.

of ARFN-MoE. The visualization of detection results on the AI-TOD dataset is presented in Fig. 6. The visualizations clearly demonstrate that our proposed method addresses the issue of missed detections that is prevalent in FFCAYOLO's predictions. Moreover, our model's results closely align with the ground truth, indicating a higher localization accuracy.

To sum up, all the experimental results show that our proposed method is very effective at detecting small objects in remote sensing images. This superior performance can be attributed to the core design of our model: the use of adaptive receptive fields. This mechanism allows the model to extract the most representative contextual features for objects of varying scales, leading to optimal detection outcomes. These results further substantiate the superiority of our model for the task of small object detection.

2) Results on VEDAI: To further validate the effectiveness of our proposed method, we conducted additional experiments on the VEDAI dataset. The experimental results are presented in Table III. As shown by the overall experimental results, our proposed ARFN-MoE model achieves 62.7% mAP_{50}, 35.7% $mAP_{50:95}$, and 34.2% mAP_s. Our ARFN-MoE model gets the best mAP_{50}, mAP_s and the second-best $mAP_{50:95}$. Even though YOLOv9m is slightly better on the $mAP_{50:95}$ score, it is only ahead by a very small margin of 0.4%. More importantly,

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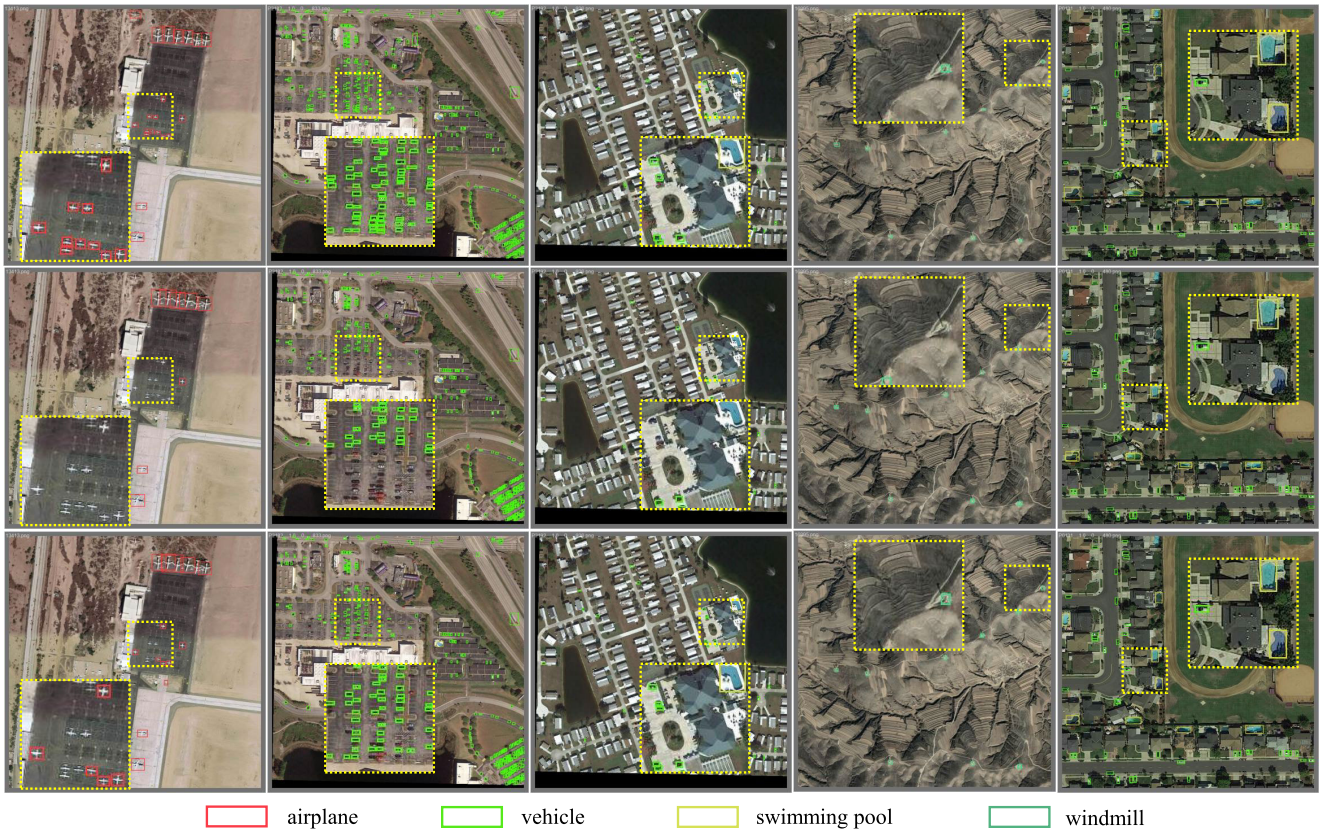


Fig. 6. Qualitative comparison of detection results between our proposed ARFN-MoE and the state-of-the-art FFCA-YOLO on the AI-TOD dataset.

TABLE III
COMPARISON OF DIFFERENT METHODS ON THE VEDAI DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Method	CA	PU	CP	TK	OT	TR	BO	VA	mAP_{50}	$mAP_{50:95}$	mAP_s	Params(M)	FLOPs(G)	FPS
FCOS [48]	63.0	45.5	54.8	21.8	36.3	19.0	8.4	10.6	32.4	16.1	15.8	32.3	51.3	44.5
Lightweight CNN [49]	74.7	56.7	56.7	36.1	26.9	56.7	22.7	36.1	52.6	-	-	-	-	-
Faster R-CNN [38]	75.6	72.0	**70.8**	**61.4**	**51.9**	60.1	**54.9**	53.1	61.8	32.6	32.7	41.8	63.6	39.7
RT-DETR [50]	69.8	63.8	61.2	60.5	49.4	52.1	44.9	46.8	56.1	33.5	32.4	32.0	103.5	29.1
YOLOv9m [51]	78.3	65.3	66.7	57.9	47.4	63.6	43.8	58.5	60.2	**36.1**	**34.2**	20.0	76.5	62.9
YOLOv10m [52]	74.5	57.8	59.5	44.4	45.9	56.0	31.7	49.8	52.5	30.9	29.3	16.5	63.5	**74.1**
YOLOv12m [53]	80.9	64.4	63.0	55.9	45.6	64.7	35.7	56.7	58.7	35.5	33.8	19.6	59.5	61.7
FFCA-YOLO [17]	81.1	75.4	66.6	52.5	44.3	63.9	**49.9**	58.8	61.6	34.9	33.5	**7.1**	**51.3**	60.9
ARFN-MoE (ours)	**81.9**	**75.7**	68.6	55.4	46.9	**67.0**	42.8	**63.0**	**62.7**	35.7	**34.2**	7.2	57.2	44.7

our model is much more efficient. It is a lot smaller, using only 36.1% of the parameters and 74.5% of the FLOPs that YOLOv9m needs. Compared with FFCA-YOLO, our proposed model achieves improvements of 1.1% and 0.8% on the mAP_{50} and $mAP_{50:95}$, respectively. The substantial performance gains over FFCA-YOLO are achieved at a very modest computational cost. Our model adds a mere 0.08-M parameters and 5.9-G FLOPs, demonstrating a highly favorable tradeoff between accuracy and complexity. We evaluated the inference speed of various models on the VEDAI testing set. Given that inference speed is susceptible to the performance and status of the hardware platform, we conducted multiple trials for each method and calculated the average values. The final average inference speeds are presented in the last column of Table III. As indicated by the results, YOLOv10m achieved the fastest inference speed of 74.1 FPS. The proposed method achieved an inference speed of 44.7 FPS, which is significantly faster than both RT-DETR and Faster-RCNN, but slightly slower than FFCA-YOLO. This marginal reduction in speed is attributed to the fact that features in our network must be processed by multiple MFE and LCE modules; this procedure introduces a minor computational overhead, thereby slightly increasing the inference time.

The visualization of detection results on the VEDAI dataset is presented in Fig. 7. Compared to other methods, our model's bounding boxes are much closer to the ground truth, which means the location is more accurate. Furthermore, our method effectively mitigates common detection failures, showing a significant reduction in missed detections (false negatives), false alarms (false positives), and misclassifications. We visualize

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Fig. 7. Qualitative comparison of detection results between our proposed ARFN-MoE and the state-of-the-art methods on the VEDAI dataset.

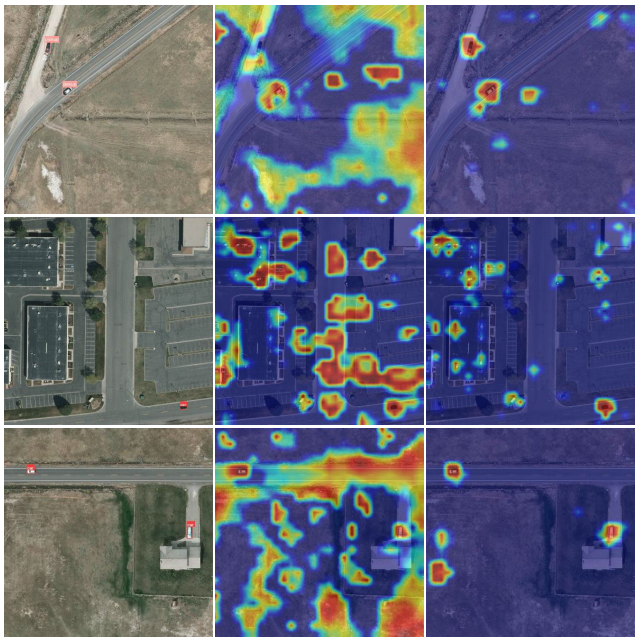


Fig. 8. Grad-CAM visualization of heatmaps generated by our proposed ARFN-MoE and the state-of-the-art FFCA-YOLO on the VEDAI dataset.

the model's heatmaps using Grad-CAM in Fig. 8. As evident from the heatmaps, compared to the comparative methods, the proposed method consistently directs stronger attention toward the small objects of interest while effectively suppressing attention in complex background regions. In particular, it significantly reduces the response to background objects that share high visual similarity with the targets, thereby mitigating the adverse effects of background clutter in remote sensing images. This superior performance is attributed to the core adaptive receptive field strategy of the proposed method. By leveraging appropriate receptive fields to capture the contextual information of small objects—rather than relying solely on their intrinsic features—our approach overcomes the detection difficulties caused by insufficient intrinsic information and eliminates interference from similar background distractors.

In summary, the collective experimental results, both quantitative and qualitative, validate the effectiveness of our proposed method. This success stems from the core design of our ARFN-MoE, which adaptively selects the optimal receptive field for each small object based on its scale. This allows the model to effectively extract both intrinsic and contextual features, overcoming the challenge of information scarcity and enabling the high-precision detection results achieved.

3) Results on DIOR: Furthermore, to verify the effectiveness of our ARFN-MoE model on general-purpose remote sensing object detection, we conducted additional comparative experiments on the DIOR dataset. The experimental results are presented in Table IV. Our proposed ARFN-MoE achieves the best overall performance with a mAP_{50} of 77.1%. This strong overall result is supported by its versatile per-class performance, where the model outperforms all competitors in 7 of the 20 categories. Standout examples include C5 (BR) at 54.6% mAP_{50}, C14 (SH) at 92.5%, and C20 (WM) at 93.1%.

The visualization of detection results on the DIOR dataset is presented in Fig. 9. The qualitative results visually demonstrate the effectiveness of our proposed method. It exhibits strong detection capabilities not only for small object classes such as AI, VE, SH, and WM, but also for classes with significant scale variations, like BR. These visual findings are in strong agreement with our quantitative results, providing further evidence for the robustness and efficacy of our approach in remote sensing small object detection.

The model's strong performance on challenging classes like SH, WM, and BR—which are characterized by small sizes and high scale variation—directly validates our adaptive receptive field approach. By tailoring the receptive field to the object's scale, our network extracts more potent and relevant

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TABLE IV
COMPARISON OF DIFFERENT METHODS ON THE DIOR DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	mAP_{50}
FSANet [43]	42.7	79.9	68.8	84.2	36.1	74.0	57.5	76.0	59.0	76.5	74.2	51.1	55.7	60.5	49.0	40.3	83.4	59.5	35.4	82.7	62.3
RetinaNet [39]	61.5	70.3	76.0	86.5	40.0	75.6	61.6	75.8	68.2	77.4	78.1	36.1	54.3	73.0	59.1	61.8	87.4	39.3	43.3	86.2	65.5
Reppoints [41]	60.5	79.1	73.4	86.2	34.8	75.8	57.1	68.2	50.6	77.2	79.6	50.4	56.2	71.7	65.5	61.3	84.4	55.4	42.3	81.0	65.5
NWD-RKA [5]	55.8	77.4	72.3	86.8	42.5	74.6	64.2	70.1	66.0	69.7	79.5	26.1	59.0	70.4	61.7	63.1	83.6	52.6	51.4	86.8	65.7
Faster R-CNN [38]	54.9	81.4	70.1	88.5	39.3	79.3	60.6	77.8	59.8	78.2	83.1	52.9	57.1	68.4	74.5	52.6	84.6	59.7	36.0	78.9	66.9
SCRDet++ [54]	64.3	79.0	73.2	85.7	45.8	76.0	68.4	79.3	68.9	77.7	77.9	56.7	62.2	70.4	67.7	60.4	80.9	63.7	44.4	84.6	69.4
MSFC-Net [55]	85.8	76.2	74.4	90.1	44.2	78.1	55.5	60.9	59.5	76.9	73.7	49.6	57.2	89.6	69.2	76.5	86.7	51.8	55.2	84.3	70.0
BSSOCNet [56]	87.4	78.4	74.5	89.1	44.3	77.6	59.8	58.5	60.2	78.2	71.8	62.4	59.2	89.7	65.7	74.0	88.5	55.6	54.4	84.8	70.7
Cascade R-CNN [40]	59.9	83.6	70.6	88.2	48.8	78.7	66.2	83.5	71.6	82.0	82.2	55.8	63.2	72.5	73.7	63.0	86.5	62.3	44.2	84.5	71.0
CFIMNet [18]	72.5	74.1	79.0	89.8	45.6	79.1	67.7	82.3	77.6	80.7	81.6	38.4	59.5	75.9	67.4	68.6	89.9	51.8	52.7	89.6	71.2
ABNet [13]	66.8	84.0	74.9	87.7	50.3	78.2	67.8	85.9	74.2	79.7	81.2	55.4	61.6	75.1	74.0	66.7	87.0	62.2	53.6	89.1	72.8
TMAFNet [14]	92.2	77.7	75.0	**91.3**	47.1	78.6	53.6	67.1	66.2	78.5	76.3	64.9	61.4	90.5	72.2	75.4	90.7	62.1	55.2	83.2	72.9
AGMFNet [57]	90.9	72.8	79.3	89.7	44.7	81.4	59.3	66.0	62.7	73.8	79.2	**65.0**	61.7	91.7	78.6	75.8	90.7	60.0	58.0	83.1	73.2
KLDet [44]	67.5	81.1	75.5	88.2	47.4	78.6	65.0	82.9	72.7	79.9	80.8	57.1	61.2	91.2	74.6	75.2	87.4	56.9	56.0	90.2	73.5
TBNet [45]	64.9	86.8	76.6	89.2	50.6	80.0	74.3	86.4	74.7	82.5	83.6	56.4	64.9	79.1	77.8	58.1	88.2	**69.4**	44.6	87.0	73.6
DetectoRS [42]	63.6	**89.9**	77.4	89.2	51.6	79.0	**75.0**	87.8	74.5	84.1	**83.7**	59.4	**65.5**	74.4	78.8	60.4	87.5	69.0	45.6	86.9	74.2
SESANet [6]	**93.5**	78.2	**91.9**	82.2	46.8	**89.3**	61.4	67.2	67.5	73.3	76.7	55.8	62.0	78.8	**84.8**	82.4	89.8	54.4	76.7	79.1	74.6
CoF-Net [58]	84.0	85.3	82.6	90.0	47.1	80.7	73.3	**89.3**	74.0	**84.5**	83.2	57.4	62.2	82.9	77.6	68.2	89.9	68.7	49.3	85.2	75.8
ARFN-MoE (ours)	85.4	75.8	83.3	**91.3**	**54.6**	79.3	57.0	83.8	**81.7**	75.9	78.8	61.1	65.1	**92.5**	79.7	**85.3**	**93.4**	59.7	65.6	**93.1**	**77.1**



Fig. 9. Visualization of detection results from our proposed method on the DIOR dataset. The figure showcases performance on several typical small object categories, such as AI, VE, and SH.

features than fixed-receptive-field methods, leading to superior detection accuracy.

E. Ablation Study

We conduct ablation experiments on the AI-TOD datasets to validate the effectiveness of each proposed component. The baseline model is FFCA-YOLO without any of the proposed components. We employ mAP_{50} to evaluate the overall performance. Moreover, we consider the impact of each component on efficiency using parameters (Params), floating point operations per second (FLOPs). The experimental results are reported in Table V.

To further validate the effectiveness of the proposed modules and the L_{aux}, we conducted multiple independent trials on the AI-TOD dataset and calculated the mean and standard deviation of the results. The average mAP_{50} scores derived from four independent trials, as presented in the table, validate the effectiveness of each individual component of the proposed method. Specifically, incorporating the MFE into the baseline network yields an average performance gain of 1.2%. On this basis, the subsequent integration of the LCE further

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TABLE V
ABLATION EXPERIMENT RESULTS OF THE PROPOSED METHOD ON THE AI-TOD DATASET

Baseline	MFE	LCE	L_{aux}	mAP_{50}	Mean	S.D.	Params(M)	FLOPs(G)
✓				62.68	62.78	0.227	7.14	81.0
✓				62.54				
✓				63.15				
✓				62.74				
✓	✓			64.09	63.98	0.298	7.20	89.1
✓	✓			64.23				
✓	✓			64.12				
✓	✓			63.47				
✓	✓	✓		64.43	64.39	0.315	7.22	92.9
✓	✓	✓		64.83				
✓	✓	✓		64.37				
✓	✓	✓		63.94				
✓	✓	✓	✓	65.32	65.19	0.226	7.22	92.9
✓	✓	✓	✓	65.11				
✓	✓	✓	✓	65.47				
✓	✓	✓	✓	64.91				

TABLE VI
EXPERIMENTS RESULTS OF MFE WITH DIFFERENT KERNELS

Expert Kernels	mAP_{50}	Params(M)	FLOPs(G)
Baseline	62.68	7.14	81.0
(5, 7, 9)	63.46	7.19	87.5
(5, 7, 9, 11)	64.09	7.20	89.1
(5, 7, 9, 11, 13)	63.96	7.22	90.1
(5, 7, 9, 11, 13, 15)	63.70	7.23	91.3

boosts performance by 0.41%. Finally, with the inclusion of the L_{aux}, the performance peaks at 65.19%, representing an additional improvement of 0.8%. The average values clearly demonstrate the efficacy of the proposed modules for small object detection. Observing the standard deviations of mAP_{50} across multiple trials in the table, it is evident that when the MFE module is used without the L_{aux}, the standard deviation approaches 0.315%. However, upon incorporating the L_{aux}, this value decreases to 0.226%. This improvement is attributed to the fact that the L_{aux} enforces a balanced selection of experts by the routing network. It effectively mitigates the load imbalance issue—where certain experts are over-utilized while others are rarely activated—thereby ensuring that all experts are adequately trained. Consequently, this leads to enhanced performance stability.

1) Evaluation of the MFE: The number of MFE plays a crucial role in the performance of multiscale small object detection. To investigate the impact of varying the number of experts, we conducted a series of experiments with different configurations. For all experiments in this analysis, the Top-k parameter of the router was consistently set to 3, and the LCE was disabled. The results presented in Table VI indicate that the optimal performance is achieved when using four experts with a kernel size step of 2. In this configuration, our model reaches a mAP_{50} of 64.09%, surpassing the baseline by 1.41%. Notably, this significant performance gain is realized with only a marginal increase in model complexity: the parameter count increases by a mere 0.06 M, and the FLOPs by only 4.8 G. We noticed that adding even more experts does not help improve the performance. This is because small objects only vary in size within a certain limit. Experts with very large receptive fields can actually hurt performance. They tend to capture too much background information, which messes with the real features of the small objects and makes the detection accuracy go down.

TABLE VII
EXPERIMENTAL RESULTS OF ROUTER USING DIFFERENT TOP-K VALUES

k	mAP_{50}	Params(M)	FLOPs(G)
1	61.48	7.20	89.1
2	63.03	7.20	89.1
3	64.09	7.20	89.1
4	63.02	7.20	89.1

The number of experts selected by the router, determined by the Top-k, also has a significant impact on detection performance. To identify the optimal Top-k value for small object detection, we conducted a series of experiments based on the model with the optimal number of experts determined previously. In this analysis, the LCE was again disabled, and all experimental settings were kept consistent, with the exception of the Top-k value. The results are presented in Table VII, increasing Top-k, the detection performance does not keep getting better. Using only one expert (Top1) is not enough because small objects have many different sizes. But using too many experts is also bad, because it adds useless or confusing information that can mess up the object's features. So, choosing the right k value is very important. Our experiments show that using three experts ($k = 3$) gives the best performance for small object detection.

Based on the preceding experimental results, we finalized the optimal configuration for our module. It consists of 4 MFE with a kernel size step of 2, and the router is set to select the Top3 experts for feature allocation. As shown in Table V, adding this MFE to the baseline model greatly improves the detection performance on the AI-TOD dataset. This significant gain is attributed to the MFE's ability to adaptively adjust the receptive field according to the scale of each small object. By leveraging its router and multiscale experts, the MFE extracts effective contextual features while avoiding the drawbacks of insufficient information for larger targets and feature redundancy for smaller ones, thereby markedly boosting overall detection performance.

2) Evaluation of the LCE: To optimize the extraction of long-range context within the LCE, we conducted experiments to determine the most effective kernel size for its horizontal and vertical depth-wise convolutions. This parameter is vital as it directly dictates the spatial range over which contextual dependencies are modeled. The comparative results for different kernel sizes are summarized in Table VIII. The result indicates that the model's performance peaks at a kernel size of 11 and degrades with larger kernels. When the kernel size exceeds 11, the network's performance begins to decline. This occurs because an excessively large kernel size possesses an overly extensive receptive field, causing the LCE to extract excessive background information. Consequently, the intrinsic features of small targets become overwhelmed by abundant background details, leading to a deterioration in the network's detection capabilities.

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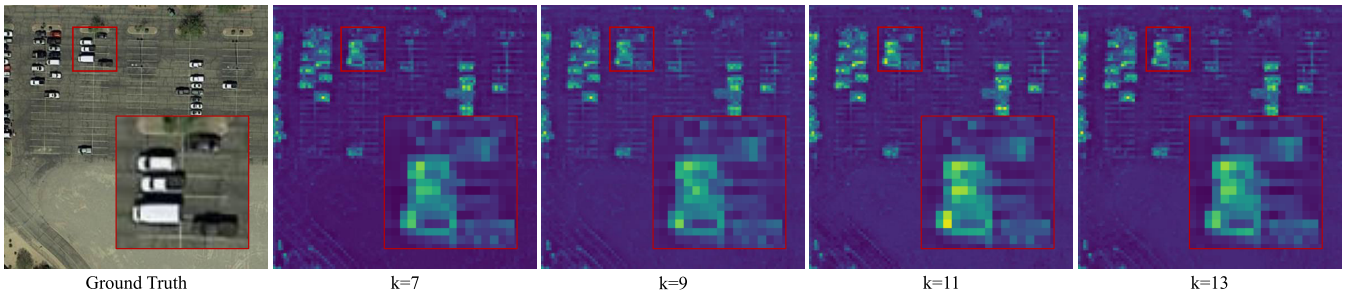


Fig. 10. Visualize the output features of LCE with different kernel sizes. It can be observed that the highest attention to small objects occurs when the kernel size is 11, at which point LCE can more effectively extract the features of small objects.

TABLE VIII
EXPERIMENTAL RESULTS OF LCE WITH DIFFERENT KERNEL SIZE

Kernel Size	mAP_{50}	Params(M)	FLOPs(G)
Baseline	62.68	7.14	81.0
7	62.87	7.15	82.9
9	63.43	7.15	83.0
11	63.99	7.15	83.2
13	63.67	7.15	83.2

We computed the channel-wise mean of the feature maps output by the LCE and visualized the results at the same spatial resolution as the original input image, as depicted in Fig. 10. As observed from the figure, the features of small objects extracted by the LCE are most prominent when the kernel size is set to 11. LCE with smaller kernel sizes yields weaker feature responses. This is attributed to the restricted receptive fields of smaller kernels, which focus primarily on the intrinsic features of the objects; given that the intrinsic features of small objects are inherently weak, the resulting representations are correspondingly faint. Conversely, excessively large kernel sizes tend to introduce substantial background information. This background noise overwhelms the target information, similarly resulting in attenuated feature extraction. The consistency between these visualization results and the quantitative ablation study further corroborates the rationality of selecting a kernel size of 11 for the LCE.

Finally, we integrated both the optimally configured LCE and the MFE module into the baseline model. As shown in Table V, this full model demonstrates a substantial performance improvement over the baseline. The primary reason for this significant gain lies in the synergistic relationship between the two modules. The LCE is responsible for capturing global, long-range contextual information for all small objects, providing a broad contextual foundation. Concurrently, the MFE employs its adaptive mechanism to extract fine-grained, scale-specific context, effectively compensating for the nuanced details that the LCE might miss. Together, they ensure the extraction of a comprehensive and highly effective set of contextual features, leading to a remarkable enhancement in small object detection performance.

3) Evaluation of the L_{Aux}: To ensure the effective training of our MoE network, we introduced an auxiliary load balancing loss. This loss is designed to encourage the router to utilize all experts effectively, thereby preventing a scenario where some experts are over-utilized while others remain inactive.

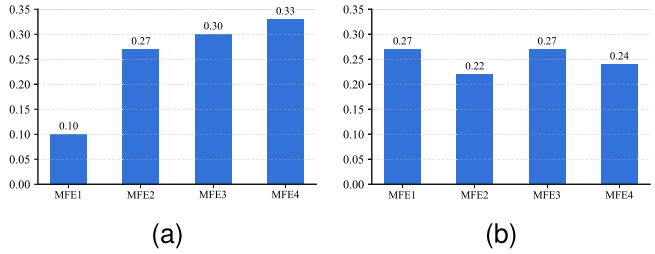


Fig. 11. Statistical analysis of the probability of selecting each expert without and without L_{aux}. (a) Without L_{aux}. (b) With L_{aux}.

The impact of incorporating this auxiliary loss is shown in the experimental results presented in Table V. The performance improvement from the auxiliary loss shows that using all experts equally is the key to getting the best results from the model. By stopping the router from favoring just a few experts, the loss makes sure every expert gets enough training. This prevents wasted parameters and helps the network learn a wider variety of useful features, which finally improves detection accuracy.

To further validate the efficacy of L_{aux}, we tallied the number of image features assigned by the router on the AI-TOD test set—both with and without L_{aux}—and calculated the corresponding assignment probabilities. The results are illustrated in Fig. 11. Fig. 11(a) depicts the assignment distribution of the model trained without L_{aux}. It is evident that the model suffers from severe expert collapse; the router disproportionately allocates image features to MFE2, MFE3, and MFE4, while MFE1 is virtually bypassed. This results in a significant workload imbalance, where MFE1 is underutilized while the other experts are over-utilized. Conversely, Fig. 11(b) presents the distribution when L_{aux} is employed. In this case, expert collapse is effectively eliminated, and the router distributes image features uniformly across all experts. Consequently, the workload achieves a balanced equilibrium, ensuring that each expert is fully utilized. These findings corroborate the effectiveness of L_{aux} in preventing expert load imbalance and facilitating the comprehensive utilization of all experts.

V. CONCLUSION

In this article, we addressed the critical limitation of the "one-size-fits-all" context extraction strategy prevalent in remote sensing small object detection. To resolve the conflict between varying object scales and fixed contextual regions, we proposed the ARFN-MoE. The primary contribution of

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this work lies in the design of scale-specialized experts combined with a grid-level router. By dynamically dispatching features to experts with optimal receptive fields, our method effectively tailors the context for each small object, mitigating the interference of background noise on extremely small targets while ensuring sufficient information for relatively larger ones. Furthermore, the introduction of the grid-level balance loss ensures the cooperative and balanced training of all experts, preventing model degradation. Extensive experiments on benchmark datasets demonstrate that ARFN-MoE significantly outperforms state-of-the-art methods, proving that adaptive context extraction is essential for robust detection in complex remote sensing environments.

For future research, we intend to investigate more advanced routing strategies, such as incorporating semantic information into the router to achieve more precise feature dispatching. In addition, we plan to validate the robustness of our adaptive framework on multimodal remote sensing data, such as combining optical and SAR images, to further enhance detection performance under adverse weather conditions.

REFERENCES

[1] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[2] T. Hoeser and C. Kuenzer, "Object detection and image segmentation with deep learning on Earth observation data: A review-part I: Evolution and recent trends," *Remote Sens.*, vol. 12, no. 10, p. 1667, May 2020.

[3] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 740–755.

[4] G. Cheng et al., "Towards large-scale small object detection: Survey and benchmarks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13467–13488, Nov. 2023.

[5] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 79–93, Aug. 2022.

[6] W. Ma, X. Wang, H. Zhu, X. Yang, X. Yi, and L. Jiao, "Significant feature elimination and sample assessment for remote sensing small Objects' detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5615115.

[7] J. Han, D. Zhang, G. Cheng, N. Liu, and D. Xu, "Object detection in high-resolution remote sensing imagery based on deep learning: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–27, 2022.

[8] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019, doi: 10.1109/TGRS.2019.2930982.

[9] Y. Gong et al., "Context-aware convolutional neural network for object detection in VHR remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 34–44, Jan. 2020, doi: 10.1109/TGRS.2019.2930246.

[10] H. Hu, S.-B. Chen, and J. Tang, "CFENet: Contextual feature enhancement network for tiny object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4703113, doi: 10.1109/TGRS.2025.3556469.

[11] G. Chen et al., "A survey of the four pillars for small object detection: Multiscale representation, contextual information, super-resolution, and region proposal," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 2, pp. 936–953, Feb. 2022.

[12] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[13] Y. Liu, Q. Li, Y. Yuan, Q. Du, and Q. Wang, "ABNet: Adaptive balanced network for multiscale object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614914, doi: 10.1109/TGRS.2021.3133956.

[14] T. Gao, Z. Liu, J. Zhang, G. Wu, and T. Chen, "A task-balanced multiscale adaptive fusion network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613515, doi: 10.1109/TGRS.2023.3289878.

[15] T. Zhang et al., "Multistage enhancement network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611512, doi: 10.1109/TGRS.2024.3363614.

[16] X. Hu et al., "SINet: A scale-insensitive convolutional neural network for fast vehicle detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 3, pp. 1010–1019, Mar. 2019.

[17] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.

[18] Z. Li et al., "Context feature integration and balanced sampling strategy for small weak object detection in remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[19] Z. Li et al., "Local to global: A sparse transformer-based small object detector for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5606516, doi: 10.1109/TGRS.2025.3525720.

[20] D. Liu, J. Zhang, Y. Qi, Y. Wu, and Y. Zhang, "Tiny object detection in remote sensing images based on object reconstruction and multiple receptive field adaptive feature enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5616213, doi: 10.1109/TGRS.2024.3381774.

[21] D. Liu, J. Zhang, Y. Qi, Y. Xi, and J. Jin, "Exploring lightweight structures for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5623215, doi: 10.1109/TGRS.2025.3567345.

[22] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Comput.*, vol. 3, no. 1, pp. 79–87, Mar. 1991.

[23] R. A. Jacobs and M. I. Jordan, "Learning piecewise control strategies in a modular neural network architecture," *IEEE Trans. Syst., Man, Cybern.*, vol. 23, no. 2, pp. 337–345, Feb. 1993.

[24] N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," 2017, *arXiv:1701.06538*.

[25] T. Chen et al., "AdaMV-MoE: Adaptive multi-task vision mixture-of-experts," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17300–17311.

[26] Y. Yang, P.-T. Jiang, Q. Hou, H. Zhang, J. Chen, and B. Li, "Multi-task dense prediction via mixture of low-rank experts," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27927–27937.

[27] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *J. Mach. Learn. Res.*, vol. 23, no. 120, pp. 1–39, 2022.

[28] J.-J. Liu, Q. Hou, M.-M. Cheng, C. Wang, and J. Feng, "Improving convolutional networks with self-calibrated convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10093–10102, doi: 10.1109/CVPR42600.2020.01011.

[29] Y. Li et al., "SM3Det: A unified model for multi-modal remote sensing object detection," 2024, *arXiv:2412.20665*.

[30] X. Cai, Q. Lai, Y. Wang, W. Wang, Z. Sun, and Y. Yao, "Poly kernel inception network for remote sensing detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27706–27716.

[31] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. TN, USA: IEEE, Jun. 2021, pp. 13708–13717, doi: 10.1109/CVPR46437.2021.01350.

[32] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[33] C. Riquelme et al., "Scaling vision with sparse mixture of experts," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2021, pp. 8583–8595.

[34] J. Wang, W. Yang, H. Guo, R. Zhang, and G.-S. Xia, "Tiny object detection in aerial images," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3791–3798.

[35] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016.

[36] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[37] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," 2021, *arXiv:2110.13389*.

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[38] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[39] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.
[40] Z. Cai and N. Vasconcelos, "Cascade R-CNN: High quality object detection and instance segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 5, pp. 1483–1498, May 2021.
[41] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9656–9665.
[42] S. Qiao, L.-C. Chen, and A. Yuille, "DetectoRS: Detecting objects with recursive feature pyramid and switchable Atrous convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10208–10219.
[43] J. Wu, Z. Pan, B. Lei, and Y. Hu, "FSANet: Feature-and-spatial-aligned network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630717, doi: 10.1109/TGRS.2022.3205052.
[44] Z. Zhou and Y. Zhu, "KLDet: Detecting tiny objects in remote sensing images via Kullback–Leibler divergence," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703316, doi: 10.1109/TGRS.2024.3382099.
[45] Z. Li, Y. Wang, D. Xu, Y. Gao, and T. Zhao, "TBNet: A texture and boundary-aware network for small weak object detection in remote-sensing imagery," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 110976, doi: 10.1016/j.patcog.2024.110976.
[46] D. Liu, J. Zhang, Y. Qi, Y. Wu, and Y. Zhang, "A tiny object detection method based on explicit semantic guidance for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: 10.1109/LGRS.2024.3374418.
[47] S. Shi, Q. Fang, X. Xu, and D. Dong, "Multiscale Gaussian attention mechanism for tiny-object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5635216, doi: 10.1109/TGRS.2025.3591260.
[48] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635, doi: 10.1109/ICCV.2019.00972.
[49] M. A. Momin, M. H. Junos, A. S. Mohd Khairuddin, and M. S. Abu Talip, "Lightweight CNN model: Automated vehicle detection in aerial images," *Signal, Image Video Process.*, vol. 17, no. 4, pp. 1209–1217, Jun. 2023.
[50] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974, doi: 10.1109/CVPR52733.2024.01605.
[51] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," 2024, *arXiv:2402.13616*.
[52] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.
[53] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.
[54] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2384–2399, Feb. 2023.
[55] T. Zhang, Y. Zhuang, G. Wang, S. Dong, H. Chen, and L. Li, "Multiscale semantic fusion-guided fractal convolutional object detection network for optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5608720, doi: 10.1109/TGRS.2021.3108476.
[56] Y. Dong, H. Yang, S. Liu, G. Gao, and C. Li, "Optical remote sensing object detection based on background separation and small object compensation strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 3341–3351, 2026, doi: 10.1109/JSTARS.2024.3351140.
[57] T. Gao, Z. Li, Y. Wen, T. Chen, Q. Niu, and Z. Liu, "Attention-free global multiscale fusion network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603214, doi: 10.1109/TGRS.2023.3346041.
[58] C. Zhang, K.-M. Lam, and Q. Wang, "CoF-Net: A progressive coarse-to-fine framework for object detection in remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600617, doi: 10.1109/TGRS.2022.3233881.



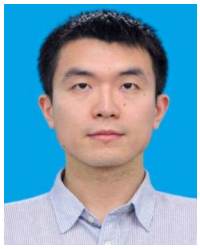
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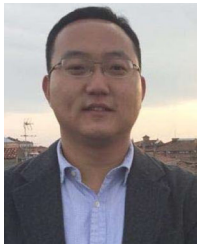
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